

Beckham Moreland

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Education

San Diego State University, College of Engineering
B.S. in Aerospace Engineering

Expected May 2029
Cumulative GPA: 3.7/4.0

Experience

Smart Mobility Intern (UAS Engineering), NASA Ames (AMA) – Moffett Field, CA Jun 2026 – Present

- Led redesign and structural validation of propulsion mounts and the avionics bay for a helicopter-deployable SAR multirotor, delivering flight-ready hardware in four days and three airframe iterations in three weeks
- Analyzed worst-case motor-thrust and rotor-wash load cases using first-principles calculations; performed FEA at $2\times$ maximum expected loads to reduce component mass by over 20% while maintaining structural margins
- Fabricated and integrated three 3D-printed airframes with nine flight-critical components using threaded inserts, soldering, and wiring; supported hardware selection and NASA procurement under NDAA constraints
- Designed a removable nose-ballast fixture and executed a hardware test plan with 10+ hover, drop, landing, and CG evaluations across four configurations; post-test inspection and PyULog analysis identified structural degradation and controller vibration, driving material, geometry, and vibration-isolation redesigns
- Evaluated six UAS concepts across cost, endurance, manufacturability, and helicopter-deployment constraints through five Air National Guard consultations, helping down-select and present a low-cost SAR multirotor for NASA and Air National Guard review

Test and Visualization Engineering Intern, MeshMap Labs Inc. – San Francisco, CA May 2025 – Sep 2025

- Led UAV and RTK field tests across four sites, varying altitude, image overlap, and capture parameters to establish repeatable data-collection procedures under changing field conditions
- Diagnosed coverage and data-quality failures through post-flight analysis, then planned corrective flights and verified revised workflows through georeferenced 3D models combining UAV imagery and terrestrial LiDAR

Full Stack Developer Intern, Axyz Technologies – Oakland, CA Jun 2023 – Jul 2025

- Developed and validated Python/Django and JavaScript features from user requirements, debugging database-backed workflows through iterative testing and stakeholder feedback

Skills

Mechanical Design & Analysis: SolidWorks, FEA, CAD assemblies, CAD drawings, first-principles load calculations

Fabrication & Integration: Additive manufacturing (3D printing), rapid prototyping, soldering, wiring, electromechanical assembly, motors, ESCs, sensors

Testing & Software: Hardware test plans, failure analysis, Python, MATLAB, pandas, Git/GitHub

Projects

Fault-Tolerant Navigation Bench, Personal Project In-Progress

- Leading end-to-end development of a bench-scale GNSS/IMU verification platform, defining the hardware architecture, MCU and sensor interfaces, logging, and test requirements for repeatable preflight validation
- Developing Python fault-injection and analysis tools to emulate GPS loss, IMU bias and noise, frozen telemetry packets, and signal dropouts, comparing detection and recovery behavior before UAV integration

Student UAV Systems Research, SDSU Undergraduate Research In-Progress

- Integrating onboard sensors, telemetry links, and ground-station controls for an experimental UAV, documenting subsystem interfaces and validating communications ahead of future hardware and flight testing
- Evaluating MATLAB-based controls and state-estimation outputs through error analysis and test documentation, identifying integration needs before implementation on the physical aircraft